



AJKD Atlas of Renal Pathology: IgA Nephropathy

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Clinical and Pathologic Features

IgA nephropathy (IgAN) presents with hematuria and varying levels proteinuria, and occurs in all age groups. The incidence of IgAN is much lower in African Americans and highest in the Asian population. The progression of disease can be rapid, especially in those with crescents. Approximately one-third of patients progress to kidney failure over time. IgAN recurs in kidney transplants in up to 60% of patients, but rarely results in graft loss. Henoch-Schönlein purpura also has mesangial IgA deposition and is morphologically similar to IgAN; however, crescents are more frequently present in biopsied Henoch-Schönlein purpura than IgAN.

Light microscopy: Variable presentation from minimal mesangial expansion to diffuse proliferation; may have sclerosis and/or crescents. Morphologic markers for worse prognosis identified by the Oxford IgAN classification include significant mesangial proliferation, any segmental sclerosis or adhesion, any endocapillary hypercellularity lesion, and tubulointerstitial fibrosis >25%.

Immunofluorescence microscopy: Dominant or codominant IgA mesangial staining with variable IgG and IgM. C3 is often present, usually equal or less than IgA staining. Deposits are polyclonal, often with λ light chains slightly more intense than κ light chains. Subendothelial deposits are often present when there is endocapillary hypercellularity.

Electron microscopy: Mesangial deposits with increased mesangial matrix and cellularity. Subendothelial deposits may also be present, particularly in more active disease with endocapillary hypercellularity.

Etiology/Pathogenesis

Antigenic galactose-deficient hinge region IgA1 plays a central role in the pathogenesis of IgAN. IgG

or IgA antibodies form immune complexes with aberrantly glycosylated IgA1 leading to glomerular deposition.

Differential Diagnosis

Morphologic clues to the diagnosis of IgA dominant postinfectious glomerulonephritis versus IgAN include endocapillary hypercellularity with neutrophils by light microscopy, C3 greater than IgA staining by immunofluorescence, and subepithelial hump-like deposits by electron microscopy.

Key Diagnostic Features

- Variable appearance on light microscopy, with mesangial proliferation to endocapillary hypercellularity. Sclerosis and/or crescents may be present.
- IgA dominant or codominant deposits shown by immunofluorescence, predominantly in the mesangium and variably in the capillary wall
- Mesangial and occasional subendothelial deposits revealed by electron microscopy

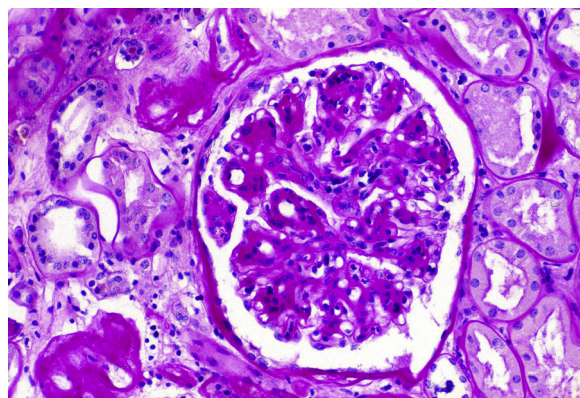


Figure 1. Immunoglobulin A nephropathy with moderate mesangial expansion and proliferation (periodic acid–Schiff stain). Reproduced with permission from *AJKD* 31(4):e1.

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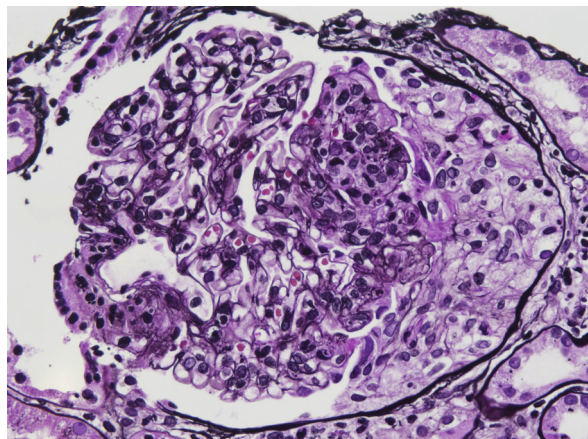


Figure 2. Immunoglobulin A nephropathy with endocapillary proliferation and cellular crescent formation (Jones silver stain).

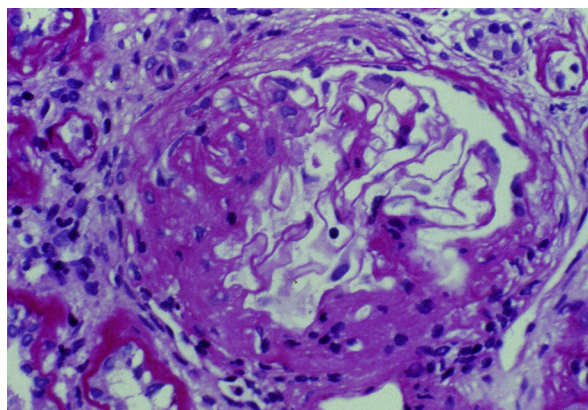


Figure 3. Immunoglobulin A nephropathy with segmental sclerosis (periodic acid–Schiff stain). Reproduced with permission from *AJKD* 31(4):e1.

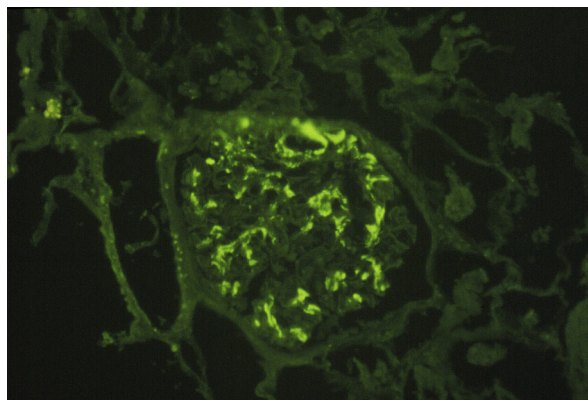


Figure 4. The diagnosis of immunoglobulin A (IgA) nephropathy is based on the finding of dominant or codominant IgA mesangial deposits. C3 is often present in the mesangium, usually equal to or less than IgA staining (immunofluorescence microscopy, IgA). Reproduced with permission from *AJKD* 31(4):e1.

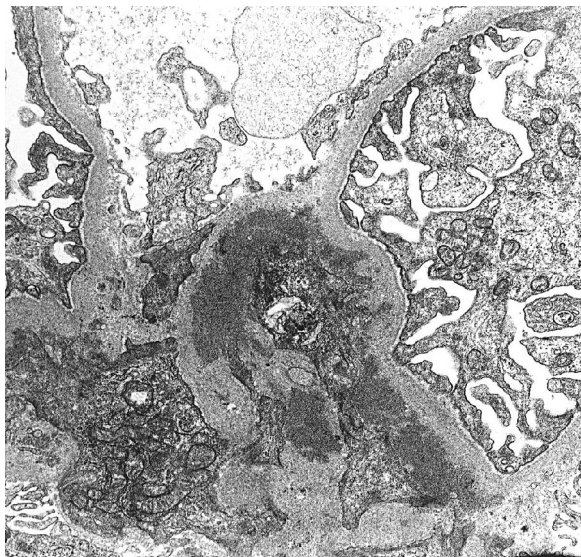


Figure 5. Mesangial deposits with increased mesangial matrix and cellularity in immunoglobulin A nephropathy (electron microscopy). Reproduced with permission from *AJKD* 31(4):e1.

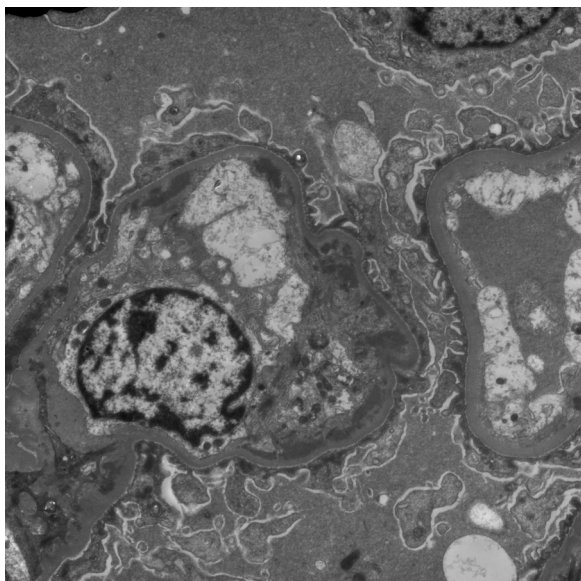


Figure 6. Subendothelial deposits in glomerular capillary walls can be seen in more active immunoglobulin A nephropathy (electron microscopy).